

## MIB (benchmark construction)

### Reference Pool



- GPT-Image-1
- 80 reference images
- 30 human + 50 non-human
- 512 × 512
- single-subject identity anchors

### Hierarchical Prompt Construction

Strict 8 → 6 → 4 → 2 subsets

36 buckets =  
3 relation types ×  
3 ratio types ×  
4 subject counts

Relation types  
(non-contact /  
occlusion /  
physical interaction)

Ratio types  
(balanced /  
human-heavy /  
object-heavy)

### Silver Set (for MIE training)

**Candidate Generation**

- NanoBanana
- MOSAIC
- 60K candidate pairs

**SOP-guided dual VLM annotation**

- Gemini-2.5 Flash
- Gemini-3.1 Flash Lite

Annotated dimensions:

- Preference
- Existence
- Appearance
- Interaction

**Agreement-based filtering**

57,055 usable pairs

Train 51K / Val 2.8K / Test 2.9K

### Gold Set (for benchmark evaluation)

**Candidate Generation**

a) 1,260 prompts → 2,520 pairs  
from four unseen generator iterations

- GPT-Image-1.5
- Flux2
- Seedream 4.5
- GLM

b) 1,500 additional pairs  
from NanoBanana vs MOSAIC vs

≈ 4K human-labeled pairs

**Double-blind human annotation**

Labels (per dimension):

- Preference
- Existence
- Appearance
- Interaction

## MIE (reference-conditioned evaluator)

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**Inputs**

Prompt: the man with green jacket is riding a bike

Reference Images:

Candidate Pair: A B

**Shared reference-conditioned representation / encoder**

**Ranking Head (pairwise preference)**

Output: Preference A > B

**Diagnosis Head (per-dimension)**

Outputs:

- Existence (binary present)
- Appearance (multi-granularity consistency)
- Interaction (yes/no/correct)

**Training (on Silver Set)**

Pairwise preference + diagnostic supervision (multitask learning)

**Evaluation (on Gold Set)**

Cross-generator evaluation

Human-aligned benchmark results

### Decoupled data regime

Distinct training (Silver) and evaluation (Gold) data for unbiased assessment.

### Scalable Silver supervision for training

60K candidate pairs, VLM SOP annotation, and agreement filtering for high-quality supervision.

### Human Gold labels for evaluation

≈4K double-blind human-labeled pairs for reliable, human-aligned benchmarking.